STEM UNIT

STORY LINE

Group 9

Title

Paper Patrol: Designing Sustainable Paper Alternatives for a Greener School

Grade Level:

Grade 6–8 (can be adjusted slightly depending on the complexity of the math and testing tasks)

Refined Real-World Problem Statement:

In many Pakistani schools, including ours, excessive use and wastage of paper contribute significantly to deforestation, pollution, and school costs. Traditional wood-based paper is made by cutting down trees, a practice that damages ecosystems and increases carbon emissions. Students will investigate how much paper their school uses and its impact, and then innovate to create sustainable paper alternatives using local or recycled materials.

Lesson	Learning Question	Learning Summary	Learning Task	Learning Outcome	Assessment	Resources Needed
1. Define the Problem	Why is paper usage a sustainability concern in our school and globally?	Students reflect on school paper usage and link it to environmental concerns.	Group discussion and mind map.	Understand environmental impact of school paper use.	Mind map submission.	Markers, whiteboard, presentation slides.
2. Learn the Science of Paper	How is traditional paper made and what are its environmental impacts?	Explore how wood-based paper is produced and its consequences.	Watch video, read text, discuss.	Explain how paper production affects the environment.	Exit slip explanation.	Video, handouts, projector.
3. Data Collection & Math	How much paper does our school use and what are the trends?	Gather real data through surveys and graph it.	Conduct surveys, create charts.	Use data to understand school paper trends.	Data tables and graphs.	Survey templates, graph tools, calculators.

Lesson	Learning Question	Learning Summary	Learning Task	Learning Outcome	Assessment	Resources Needed
4. Explore Stone Paper	What is stone paper and how might it be a better alternative?	Learn what stone paper is, its ingredients and environmental benefits.	Research and group discussion.	Describe components and benefits of stone paper.	Group presentation.	Articles, internet, chart paper.
5. Design a Prototype	How can we create stone paper using local materials?	Plan out the method and materials to use in the prototype.	Write procedure and prep materials.	Develop a step-by-step prototype plan.	Written procedure.	Planning sheet, sample materials.
6. Try a Solution (Prototype 1)	Can our planned method produce a paper-like material?	Create and dry the first stone paper version.	Mix, roll, mold and dry material.	Produce and observe first prototype.	Teacher observation/photos.	Calcium carbonate, resin, molds, gloves.
7. Test the Solution	How does our prototype compare to regular paper?	Test the product for key characteristics.	Perform strength, writability, and water tests.	Evaluate prototype performance.	Test report completion.	Rulers, weights, water, pens.
8. Analyze & Redesign	What improvements are needed in our prototype?	Review test data and identify improvement areas.	Analyze, redesign plan.	Use engineering cycle to improve design.	Redesign notes.	Test results, planning sheets.
9. Final Prototype Build	Can we improve our stone paper with our new plan?	Build a final prototype using revised design.	Construct and cure improved product.	Implement feedback and refine solution.	Peer review.	Materials from earlier + any extras.
10. Communicate the Solution	How can we share our findings with our school community?	Prepare and present findings in a showcase.	Make digital or poster presentation.	Effectively communicate STEM solution.	Rubric-based evaluation.	Laptops, chart paper, projector.

Summary

Science Involved

- **Understanding traditional paper production**: materials, energy use, and environmental effects.
- **Exploring alternative materials**: calcium carbonate and resin as eco-friendly replacements.
- **Environmental science**: impact of deforestation, waste accumulation, and carbon footprint.
- **Material science**: properties of stone paper—strength, flexibility, water resistance, biodegradability.

Math Involved

- **Data collection and statistics**: surveys on paper usage, calculating averages (mean, median, mode).
- **Graphing and representation**: creating bar graphs, pie charts, line graphs from real data.
- **Cost analysis**: comparing school paper costs with alternative materials.
- **Mathematical modeling**: estimating trees saved, projecting future paper savings, trend analysis.
- **Ratio and proportion**: calculating material mixes for prototype development.

Engineering Involved

- **Engineering Design Process**:
    - Define the problem
    - Plan a solution
    - Build and test a prototype
    - Analyze and redesign
    - Communicate results

- **Prototyping**: creating a functional sample of stone paper using local materials.
- **Testing and evaluation**: measuring performance (foldability, strength, writability, water resistance).
- **Iteration**: improving the prototype based on test results.

Integration of Stone Paper into the Engineering Design Process

EDP Stage	Student Action	Use of Stone Paper & Calcium Carbonate
1. Define the Problem	Identify excessive paper use and its environmental impact.	Introduce the idea of non-wood alternatives like stone paper. Frame the problem around replacing traditional paper.
2. Learn About the Problem	Research how paper is made and its ecological cost.	Explore stone paper: made from calcium carbonate (limestone) and resin. Compare environmental benefits to wood pulp.
3. Plan a Solution	Brainstorm and outline a prototype.	Students plan how to mix calcium carbonate powder with a small amount of resin. Consider sourcing, safety, and tools needed.
4. Create a Prototype	Make a small sheet of paper-like material.	Mix calcium carbonate and binder, mold it into sheets, and dry it. This is their first trial at making stone paper.
5. Test the Prototype	Evaluate based on criteria: writability, foldability, water resistance.	Students compare their stone paper to traditional paper using performance tests. Document test results.
6. Analyze Results & Redesign	Review test data and find ways to improve.	Adjust the ratio of calcium carbonate to resin, improve molding/drying techniques, or add surface treatments.
7. Final Prototype	Build a revised, improved version.	Create a more refined version of stone paper that performs better across the selected criteria.

EDP Stage	Student Action	Use of Stone Paper & Calcium Carbonate
8. Communicate Solution	Share process and results through a showcase.	Present the benefits of their stone paper—cost, environmental impact, and performance compared to wood-based paper.

Why Calcium Carbonate?

- **Abundant and eco-friendly**: Requires no trees, uses less water and energy.
- **Durable and recyclable**: Stone paper is tear-resistant and water-resistant.
- **Innovative application**: Makes students feel like real-world eco-engineers solving sustainability problems.

Technology Involved

- **Digital tools for presentations**: using slides, posters, and data visuals.
- **Online research**: investigating stone paper production, sustainability practices.
- **Data recording**: spreadsheets, graphing software for trends and statistics.
- **Presentation and communication**: using tools like Google Slides or Canva for final sharing.